
Monetary Policy Shocks and Macro-Financial Dynamics: A Time-Series Analysis

Time-Series Project at the Department of Econometrics

Applied Econometrics & Forecasting

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List of Abbreviations

ACF	Autocorrelation Function
ADF	Augmented Dickey–Fuller test
BIC	Bayesian Information Criterion
CPI	Consumer Price Index
DE1Y	German 1-year government bond yield
EA-MPD	Euro Area Monetary Policy Event-Study Database
ECB	European Central Bank
HAC	Heteroskedasticity and Autocorrelation Consistent
HICP	Harmonised Index of Consumer Prices
KPSS	Kwiatkowski–Phillips–Schmidt–Shin test
MP	Monetary Policy
MPS	Monetary Policy Shock
OIS	Overnight Index Swap
PACF	Partial Autocorrelation Function
PC	Principal Component
PCA	Principal Component Analysis
VSTOXX	Euro Stoxx 50 Volatility Index

1 Introduction

Monetary policy plays a central role in shaping macroeconomic and financial conditions in the euro area. Understanding how financial markets and macroeconomic variables respond to unexpected monetary policy decisions is important for academic research and for policy evaluation. In particular, high-frequency identification around central bank announcements has become a popular approach to recover monetary policy surprises, as financial markets react instantaneously to new information released by the central bank.

This project follows the methodology of Altavilla et al. (2019) and Bauer and Swanson (2023) and aims to construct a monetary policy shock (MPS) measure for the euro area using high-frequency movements in Overnight Index Swap (OIS) rates around ECB press releases. The shocks are extracted using principal component analysis (PCA) applied to OIS changes at different maturities. These surprises are then converted into a monthly series by projecting them onto macro-financial variables before the ECB announcements. The resulting monthly shock serves as a proxy for unexpected monetary policy changes.

The central research question of this study is how Luxembourg consumer prices respond to these monetary policy shocks. Luxembourg is a small and highly open economy within the euro area. To analyse these dynamics, we employ a direct forecasting approach based on Local Projections (Jordà, 2005), combined with LASSO-based model selection using information criteria and heteroskedasticity-and-autocorrelation robust (HAC) inference.

The dataset combines high-frequency OIS data around ECB monetary policy events with monthly macroeconomic data for Luxembourg, including Luxembourg CPI (the variable of interest) and the German 1-year government bond yield. The resulting series are examined for stationarity, transformed where necessary and used into the forecasting regression framework.

The paper is structured as follows. Section 2 describes the data and presents the exploratory analysis. Section 3 outlines the methodological framework, including PCA shock identification, stationarity testing, variable transformations and the LASSO-based local projection estimation strategy. Section 4 presents the empirical results and the last section concludes.

2 Data Description

This section describes the dataset used to construct the monetary policy shock (MPS) and to estimate the dynamic response of Luxembourg inflation. The analysis combines financial market data around ECB monetary policy announcements with monthly macroeconomic indicators from June 2003 to October 2023. All figures referenced in this section are provided in the Appendix.

2.1 High-Frequency OIS Data

The MPS is constructed using intraday OIS data from the EA-MPD dataset, following the identification approaches of Altavilla et al. (2019) and Bauer and Swanson (2023). The EA-MPD provides OIS rate changes around ECB Monetary Policy events. For each event, we use OIS maturities from the very short end to the two-year segment,

SW, 1M, 3M, 6M, 1Y, 2Y,

We focus on the Press Release Window, which isolates market reactions in a tight 30-minute interval surrounding the publication of the monetary policy statement. The rate changes serve as inputs to the PCA following Bauer and Swanson (2023).

2.2 Monthly Aggregation and Macro-Financial Variables

To obtain a monthly monetary policy shock, the PCA surprise from the Press Release Window is aggregated at monthly frequency. Following Bauer and Swanson (2023), the resulting monthly surprise is cleaned from any predictable component by projecting it on macro-financial variables observed prior to the policy announcement.

The following variables are included in our model:

- STOXX Europe 600 Index (source: ECB Data Portal),
- VSTOXX volatility index (source: ECB Data Portal),
- Euro Area Industrial Production (source: ECB Data Portal),
- U.S. S&P 500 Index (source: Yahoo Finance),
- Euro Area HICP (source: Eurostat).

These are standard predictors of financial conditions and macroeconomic sentiment. Figures 1 - 4 show that most of these series exhibit clear upward trends, volatility clustering, or structural breaks, indicating non-stationarity. Their formal transformations and stationarity adjustments are discussed in the Methodology section.

2.3 Macroeconomic Variables for the local projection

The final part of the empirical analysis uses two monthly macro-financial variables: the Luxembourg consumer price index (CPI) and the German 1-year government bond yield (DE1Y). The CPI series is obtained from STATEC Luxembourg, while the DE1Y yield is taken from the EA-MPD dataset. Both series are available at monthly frequency from June 2003 to October 2023.

Figure 4 (a) shows the Luxembourg CPI level. The series displays a smooth and persistent upward trend throughout the sample, reflecting the long-run inflation pattern in Luxembourg. The trend steepens notably after 2020, consistent with the strong inflation throughout the euro area following the COVID-19 shock. The clear trending behavior indicates non-stationarity. Figure 4 (b) presents the DE1Y series. Unlike the CPI, the 1-year government bond yield fluctuates around zero with occasional sharp spikes, reflecting periods of financial stress and changing monetary policy expectations. Spikes are observed around the Global Financial Crisis, the euro area sovereign debt crisis, and the pandemic period. The absence of a persistent trend and the mean-reverting behavior suggest that the DE1Y series is stationary.

These two variables form the core dataset used in the local projection framework to estimate the dynamic response of Luxembourg inflation to the monetary policy shocks.

3 Methodology

This section describes the empirical strategy used to construct the monetary policy shock series and to estimate the dynamic effect of euro area monetary policy on Luxembourg inflation. The implementation closely follows the high-frequency identification approach of Altavilla et al. (2019) and the predictability-correction and local projection framework in Bauer and Swanson (2023).

The empirical analysis proceeds in three steps: (i) extracting a high-frequency monetary policy factor from OIS rates using principal component analysis (PCA),

(ii) aggregating this factor to monthly frequency and purging it of predictable components via macro-financial regressions, and (iii) estimating impulse responses of Luxembourg inflation using local projections with post-double-selection LASSO and heteroskedasticity-autocorrelation robust (HAC) standard errors.

3.1 Extraction of Monetary Policy Shocks

To identify the monetary policy shocks this study follows the approach introduced by Gürkaynak et al. (2005), adapted to the Euro Area by Brand et al. (2010) and incorporated into the EA-MPD methodology of Altavilla et al. (2019). The central idea of this literature is that yield changes around monetary policy announcements are driven by a small number of latent factors, which capture the unexpected component of monetary policy decisions.

Let X_t denote the matrix of intraday OIS yield changes in the press release window at policy date t . Each row corresponds to a monetary policy event, and each column represents the change in an OIS rate of a specific maturity.

We assume that these yield changes follow this factor structure,

$$X_t = F_t \Lambda' + \varepsilon_t, \quad (1)$$

where F_t contains the latent common factors, Λ is the loading matrix, and ε_t represents idiosyncratic movement. The factors are estimated with principal component analysis, which captures the directions along which the OIS rates co-move the most.

Gürkaynak et al. (2005) show that these factors represent the main unexpected policy changes that financial markets react to. Brand et al. (2010) demonstrate that this framework applies similarly to the Euro Area, with distinct factors emerging in the press release and press conference windows. Altavilla et al. (2019) show that, in the press release window, a single dominant factor typically characterizes unexpected monetary policy actions.

In line with this literature, we use the first principal component of X_t as the monetary policy shock. We then aggregate it to monthly frequency and use it as baseline for the following steps of this study.

3.2 Predictability of the Monthly Monetary Policy Shock

The monthly monetary policy surprise obtained from the PCA may still contain predictable variation driven by macro-financial conditions. Bauer and Swanson (2022)

find that U.S. variables like employment growth and stock market movements contain predictive power for monetary policy surprises. We now adapt this idea to the euro area. If such predictability is not removed, the resulting series cannot be interpreted as an exogenous monetary policy shock. Following their approach, we regress the monthly factor on a set of lagged macro-financial indicators to isolate the unpredictable component. Let mps_t denote the monthly monetary policy factor and let z_{t-1} be a vector of macro-financial variables observed prior to month t . z_{t-1} includes transformations of the STOXX index, the VSTOXX volatility index, euro area industrial production, euro area inflation and the S&P 500.

Before executing the regression, each predictor is transformed to achieve stationarity. This is important because the presence of unit roots in the regressors may lead to spurious regression (Granger and Newbold, 1974). To address this issue, we examine the time-series properties of all predictors using Augmented Dickey–Fuller (ADF) and KPSS tests (Dickey and Fuller, 1979; Kwiatkowski et al., 1992). Whenever the level series exhibits non-stationarity, we apply log-differences or multi-period differences to obtain a stationary transformation. This ensures that the subsequent regression is well specified. The testing outcomes themselves are presented in the results section.

The predictability regression takes the form

$$mps_t = \alpha + \gamma' z_{t-1} + u_t, \quad (2)$$

where u_t represents the unpredictable component of the monthly monetary policy surprise. The use of overlapping multi-month changes in several predictors naturally induces serial correlation and heteroskedasticity in the regression errors. To obtain valid inference, we therefore report heteroskedasticity- and autocorrelation-consistent standard errors.

The cleaned monetary policy shock is defined as the regression residual,

$$mps_t^c = \hat{u}_t, \quad (3)$$

which isolates the component of the monthly monetary policy factor that cannot be forecast from information available prior to the policy meeting. This series serves as the exogenous monetary policy shock in the following part.

3.3 Local Projections

The final part of the analysis examines the dynamic response of Luxembourg consumer price index (CPI) to the cleaned monetary policy shock. Following the framework described in Bauer and Swanson (2022), we estimate a sequence of horizon-specific regressions of the form

$$y_{2,t+h} = \beta_0^{(h)} + \beta_1^{(h)} z_t + \beta_2^{(h)} x_{t-1} + u_{t+h}, \quad (4)$$

where z_t denotes the monetary policy shock, $y_{2,t}$ is the monthly log change in Luxembourg CPI, $y_{1,t}$ is the 1-year interest rate, and $h = 0, 1, \dots, 16$ corresponds to the forecast horizon. The vector x_{t-1} contains lagged values of the shock, the interest rate, and the inflation series. The parameter of interest is $\beta_1^{(h)}$, which represents the impulse response of inflation to a monetary policy shock.

Before estimating these regressions, the time series used in the model are examined for stationarity. This is essential to avoid spurious regression issues in the local projection. We apply again standard unit root tests, including the Augmented Dickey–Fuller (ADF) test and the KPSS test. If a time series is non-stationary, we apply appropriate transformations such as log-differences.

Model selection for the set of controls is conducted using the post-double-selection LASSO procedure used by Belloni et al (2014). This method selects control variables by running two separate LASSO regressions: one for the MPS and one for $y_{2,t}$ estimating the final OLS equation using the union of selected controls. This approach ensures valid inference in high-dimensional settings and has become standard in empirical macroeconomic and financial applications. The final post-LASSO OLS regression for horizon h is therefore

$$y_{2,t+h} = \beta_0^{(h)} + \beta_1^{(h)} z_t + \theta^{(h)'} \tilde{x}_{t-1} + u_{t+h}. \quad (5)$$

Because of overlapping lags, the resulting errors may be autocorrelated and heteroskedastic. For this reason, all inference is based on heteroskedasticity- and autocorrelation-consistent (HAC) standard errors. These are obtained using a Newey–West estimator. For each horizon h , we report the point estimate of $\beta_1^{(h)}$ and its 95 percent HAC confidence interval.

Finally we do standard diagnostic checks for the regression residuals, including ACF/PACF plots, Ljung–Box and Breusch–Godfrey tests for autocorrelation, and the Breusch–Pagan test for heteroskedasticity. These diagnostics confirm whether

the HAC corrections are appropriate.

4 Results

4.1 PCA Results

This section presents the results of the principal component analysis applied to the OIS rate changes around ECB press release announcements. Table 1 in the Appendix reports the variance decomposition of the PCA. The first principal component explains approximately 75% of the total variation, while the second component accounts for an additional 18%. The remaining components contribute less than 5% each and mainly capture noise. This pattern indicates that a single dominant factor drives the co-movement of OIS rates, consistent with the findings of Gürkaynak et al. (2005) for the United States and Brand et al. (2010) for the Euro area.

The loadings of the first principal component, reported in Table 2 in the Appendix, are positive across all maturities and larger for short- and medium-term OIS rates.

Figure 5 plots the first principal component over the sample period. The series shows sharp spikes that happen during important episodes of monetary and financial stress, including the global financial crisis, the sovereign debt crisis, and the COVID-19 pandemic period. These patterns demonstrate that the extracted factor captures the immediate market reaction to ECB monetary policy announcements.

4.2 Time-Series Properties and Stationarity Tests

Before analysing the predictability of the MPS, we first examine the time-series properties of the macro-financial predictor variables. Table 3 shows that, with the exception of the German 1-year bond yield (DE1Y), the macro-financial predictors are non-stationary: the ADF test typically fails to reject a unit root, while the KPSS test rejects covariance stationarity. Figures 6 and 7 illustrate this behaviour for the Luxembourg CPI series, the CPI shows a slowly decaying ACF, and a PACF that features a very large spike at the first lag. Even though the ACF and PACF for the CPI log-difference indicates dependencies between the lags, the unit-root tests in Table 4 classify the series as stationary (ADF strongly rejects a unit root and the KPSS test does not reject stationarity at the 5% level). To avoid spurious regression and invalid inference, the non-stationary variables are transformed to

growth rates or log changes, as summarised in Table 4. In contrast, the constructed monthly monetary policy shock in Table 4 already behaves as a stationary series, and Figures 6 and 7 show only short-lived autocorrelation. After transformation, the remaining variables pass both ADF and KPSS checks (with at most borderline evidence for the S&P 500), and their ACFs and PACFs decay rapidly, so that all regressors used in the subsequent analysis can be treated as stationary.

4.3 Predictability of the Monthly Monetary Policy Shock

Table 5 compares alternative specifications for the MPS regression. Both the AIC and BIC are lowest for Model 4, indicating a modest but clear improvement in fit relative to the simpler specifications, so we focus on this model. Model diagnostics for Model 4, summarised in Table 6, point to moderate multicollinearity (all VIF-type measures remain well below conventional thresholds) and no evidence of positive serial correlation (Durbin-Watson statistic close to 2.4), but a highly significant Breusch-Pagan test and strong rejection of normality in the Shapiro-Wilk test. These results indicate pronounced heteroskedasticity and non-normal residuals, so inference is based on HAC-robust (Newey-West) standard errors.

Table 7 reports the coefficients of Model 4 together with HAC-robust standard errors. The adjusted R^2 of about 0.12 confirms that only a relatively small share of the variation in the MPS is predictable from macro-financial conditions. In the preferred specification, the quadratic term in the three-month STOXX return is statistically significant at the 5% level, suggesting that large equity price movements are associated with larger monetary policy surprises. The quadratic term in VSTOXX and the 15-month HICP change are only significant at the 10% level, while the linear effects of STOXX, VSTOXX, S&P 500 returns and industrial production growth are not statistically different from zero. The cleaned monetary policy shock used in the subsequent analysis is therefore given by the residuals from Model 4.

4.4 Local projection results

Figure 8 plots the estimated impulse response of Luxembourg CPI inflation to the cleaned monetary policy shock together with 95% Newey-West confidence bands, and Table 8 reports the corresponding coefficients and confidence intervals. The estimated responses are generally small in magnitude and imprecisely estimated. For horizons $h = 0$ to $h = 6$, the point estimates of $\hat{\beta}_1^{(h)}$ fluctuate around zero and the 95% confidence intervals always include zero, indicating no statistically

significant effect of the monetary policy shock on monthly inflation. At horizon $h = 7$, the response becomes negative and statistically significant at the 5% level: the coefficient is about -0.11 , with a 95% confidence interval from roughly -0.21 to -0.00 . For longer horizons ($h \geq 8$), the coefficients again fluctuate around zero and their confidence intervals remain wide and include zero, so there is no robust evidence of persistent longer-run effects.

Table 9 reports diagnostic tests for the residuals of the baseline local projection. Both the Ljung-Box and Breusch-Godfrey tests provide clear evidence of residual serial correlation, while the Breusch-Pagan test points to heteroskedasticity. Taken together, these results indicate that the errors deviate from the classical OLS assumptions and motivate the use of HAC-robust (Newey-West) standard errors throughout.

5 Conclusions

The empirical results show how Euro area monetary policy surprises are perceived by financial markets and how they transmit, if at all, to inflation in a small member state. The PCA of OIS rate changes shows that a single factor accounts for strong movements with positive loadings. This confirms the idea described by Gürkaynak et al. (2005) and Brand et al. (2010). The spikes in the extracted factor around the global financial crisis, the Euro area sovereign debt crisis and the COVID-19 period indicate that policy communication during those episodes contained particularly strong, and often surprising, information for markets.

The second part of our analysis shows that only a few predictors are statistically relevant, and in a non-linear way. The key result is the significant positive coefficient on the quadratic term in three-month STOXX returns, whereas the linear STOXX term is insignificant. This implies that small or moderate stock market movements are not systematically associated with policy surprises, but *very large* equity price changes have an impact on MPS. In periods of market rallies or sharp corrections, monetary policy surprises become larger. A similar but weaker pattern is shown for the squared VSTOXX term, suggesting that episodes of high uncertainty are also associated with larger policy surprises. Overall, the pattern of significance points to ECB surprises being particularly large in extreme market environments, rather than linearly to normal fluctuations.

The impulse response estimates suggest that unexpected ECB monetary policy shocks have only a very limited impact on Luxembourg inflation. There is no sta-

tistically detectable effect on monthly CPI inflation in the first six months after a shock, and even the significant response at a horizon of seven months points only to a modest and temporary disinflation. Beyond this short window, the effects disappear again, indicating that in a small and highly open economy like Luxembourg, domestic inflation is driven primarily by other factors rather than by Euro area monetary policy surprises.

A Appendix

A.1 Tables

A.1.1 Principal Component Analysis

Table 1: PCA Variance Decomposition

	PC1	PC2	PC3	PC4	PC5	PC6
Standard deviation	2.1220	1.0420	0.4936	0.3509	0.1551	0.1428
Proportion of variance	0.7505	0.1810	0.0406	0.0205	0.0040	0.0034
Cumulative proportion	0.7505	0.9315	0.9721	0.9926	0.9966	1.0000

Table 2: PC1 Loadings

	OIS_SW	OIS_1M	OIS_3M	OIS_6M	OIS_1Y	OIS_2Y
Loading	0.379	0.404	0.446	0.458	0.419	0.330

A.1.2 Unit Root and Stationarity Tests

Table 3: ADF and KPSS Tests for Macro-Financial Variables (Levels)

Variable	ADF rejects unit root?	KPSS rejects stationarity?	Conclusion
STOXX price index	No	Yes	Non-stationary
VSTOXX index	Yes	Yes	Effectively non-stationary
HICP (EA)	No	Yes	Non-stationary
Industrial production (EA)	Yes	Yes	Non-stationary
S&P 500 price	No	Yes	Non-stationary
Luxembourg CPI (level)	No	Yes	Non-stationary
German 1Y bond yield (DE1Y)	Yes	No	Stationary

Table 4: ADF and KPSS Tests for Transformed and Constructed Variables

Variable	ADF rejects unit root?	KPSS rejects stationarity?	Conclusion
STOXX 3M log change	Yes	No	Stationary
VSTOXX 3M log change	Yes	No	Stationary
HICP 15-month change	Yes	No	Stationary
Industrial production growth	Yes	No	Stationary
S&P 500 3M log change	Yes	Yes [†]	Stationary (borderline)
Luxembourg CPI (log-difference)	Yes	No [‡]	Stationary
Monthly MP shock	Yes	No	Stationary

Notes: [†] KPSS rejects stationarity at the 5% level; [‡] KPSS does not reject stationarity at the 5% level.

A.1.3 MPS Regression and Diagnostics

Table 5: Model Comparison: AIC and BIC

Model	AIC	BIC	N
Model 1	583.69	607.05	208
Model 2	581.83	611.87	208
Model 3	573.84	613.89	208
Model 4	573.50	606.88	208

Table 6: Diagnostic tests for the preferred MPS specification (Model 4)

Test	Statistic	p-value	Conclusion
Max. VIF ($\text{GVIF}^{1/(2 \cdot \text{Df})}$)	2.34	—	Moderate multicollinearity
Durbin–Watson	2.36	0.99	No evidence of positive autocorrelation
Breusch–Pagan (studentised)	47.73	0.00	Heteroskedasticity present
Shapiro–Wilk	0.72	0.00	Residuals not normally distributed

Table 7: Regression results for the monthly MPS (Model 4)

Variable	Coefficient	HAC s.e.
Intercept	-0.217***	(0.083)
STOXX 3M	1.360	(1.698)
STOXX 3M ²	32.187**	(14.341)
VSTOXX 3M	0.252	(0.338)
VSTOXX 3M ²	1.958*	(1.133)
S&P 500 3M log change	-1.034	(1.743)
HICP 15-month change	0.175*	(0.105)
Industrial production growth	0.022	(0.058)
STOXX 3M $\times$ VSTOXX 3M	7.008	(9.662)
Observations	208	
R^2	0.157	
Adjusted R^2	0.123	
Residual s.e.	0.936	
AIC	573.50	
BIC	606.88	

Notes: The dependent variable is the monthly MPS. The table reports OLS coefficients; HAC-robust (Newey–West) standard errors are shown in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

A.1.4 Local Projections and Inflation Response

Table 8: Local projection estimates of the response of Luxembourg inflation to the MPS

Horizon h	$\hat{\beta}_1^{(h)}$	95% lower	95% upper
0	-0.066	-0.170	0.038
1	-0.047	-0.161	0.067
2	0.056	-0.010	0.123
3	-0.033	-0.137	0.071
4	0.031	-0.069	0.132
5	0.014	-0.059	0.087
6	0.045	-0.041	0.130
7	-0.107**	-0.210	-0.004
8	0.054	-0.060	0.169
9	-0.014	-0.145	0.117
10	-0.016	-0.102	0.069
11	0.034	-0.071	0.139
12	0.033	-0.087	0.154
13	-0.031	-0.162	0.100
14	-0.014	-0.183	0.155
15	-0.080	-0.224	0.063
16	0.019	-0.076	0.114

Notes: The table reports horizon-specific local projection estimates of the response of monthly Luxembourg CPI inflation to the cleaned monetary policy shock. Coefficients are obtained from post-double-selection local projections; 95% confidence intervals are based on Newey-West HAC standard errors. **, and *** denote significance at the 5% and 1% level, respectively (a single * would indicate 10% significance). Only horizon $h = 7$ is statistically significant at the 5% level.

Table 9: Diagnostic tests for local projection residuals

Test	Statistic	df	p-value
Ljung-Box (lag 10)	82.83	10	< 0.0001
Breusch-Godfrey (order 2)	7.16	2	0.028
Breusch-Pagan	11.94	2	0.0026

Notes: Tests are based on the residuals of the baseline local projection for $h = 0$. The Ljung-Box and Breusch-Godfrey tests both reject the null of no serial correlation, and the Breusch-Pagan test rejects homoskedasticity, motivating the use of HAC-robust (Newey-West) standard errors for all horizons.

A.2 Figures

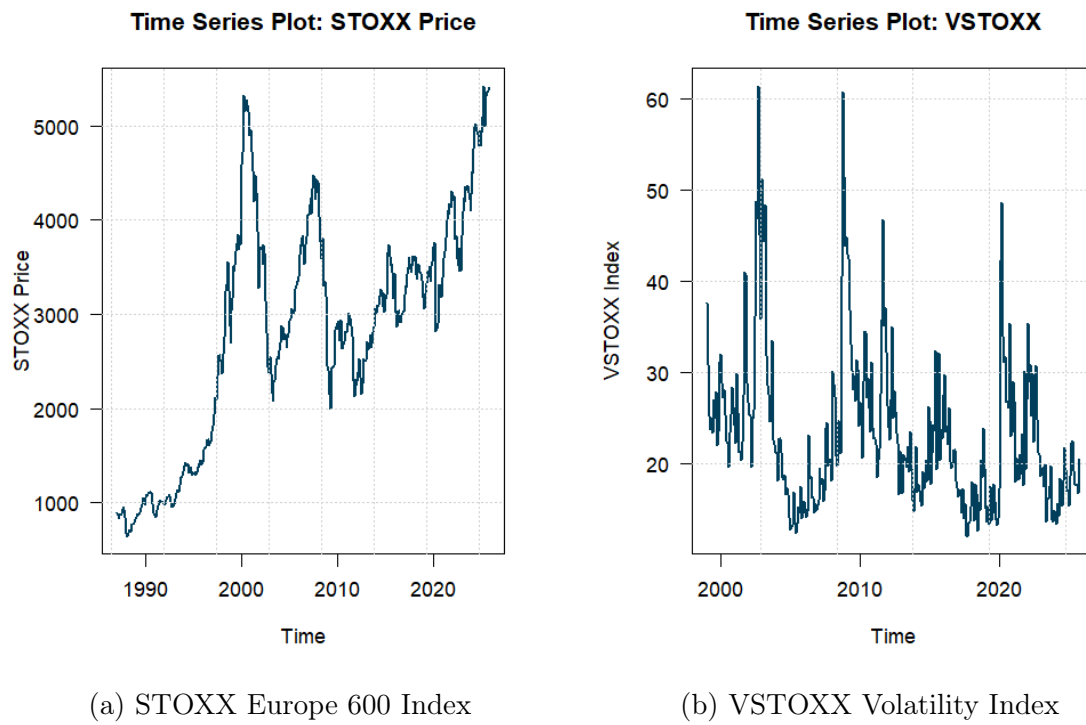


Figure 1: Equity Prices and Volatility

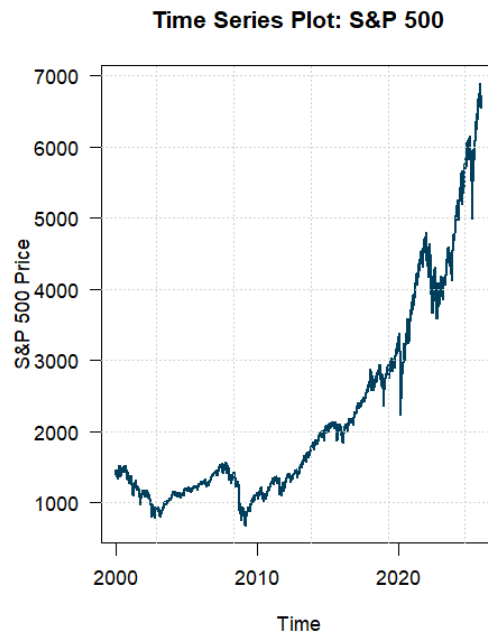
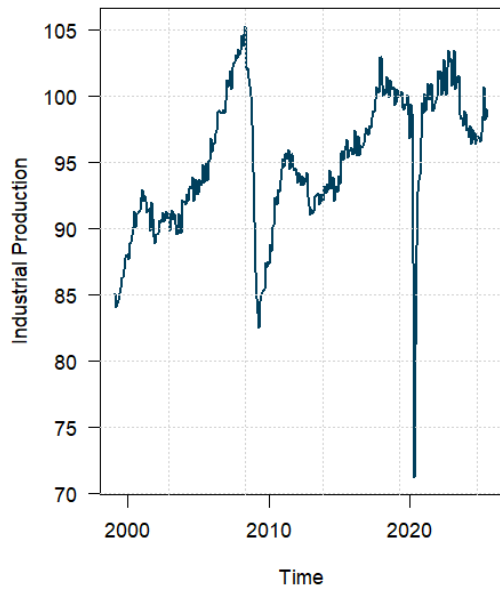


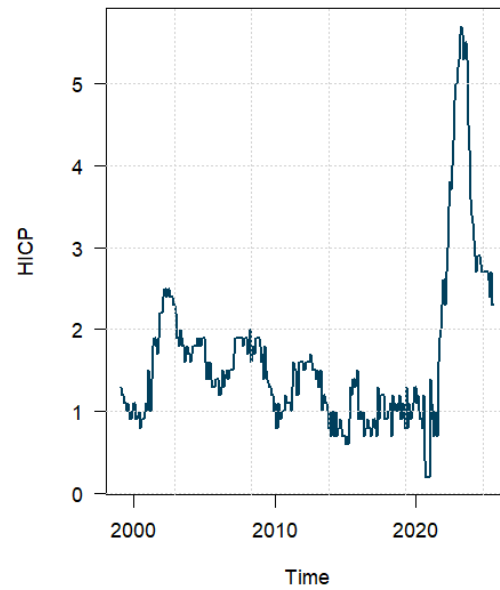
Figure 2: S&P 500 Index

Time Series Plot: Industrial Production (EA)



(a) Euro Area Industrial Production

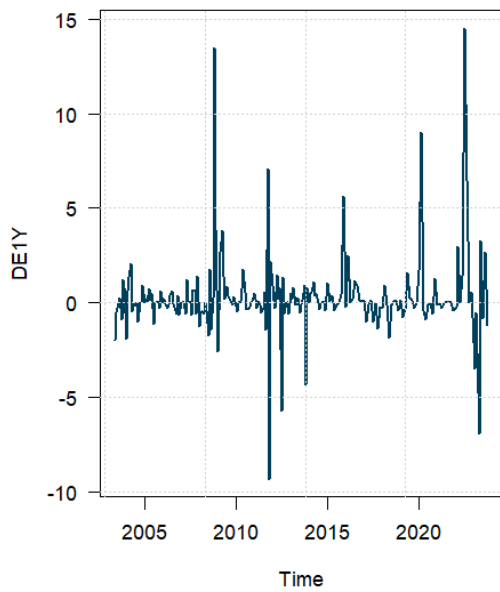
Time Series Plot: HICP



(b) Euro Area HICP

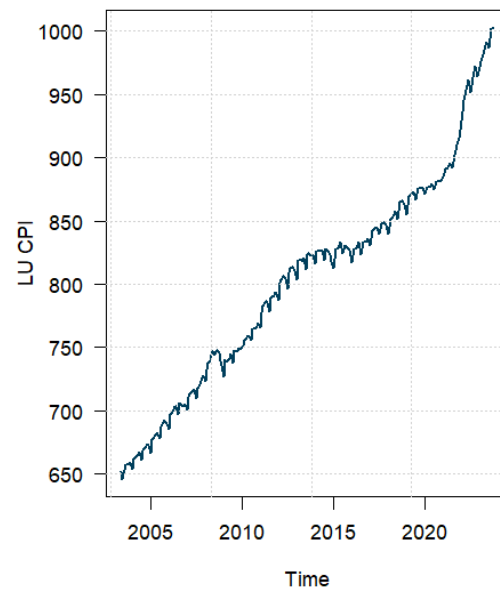
Figure 3: Euro Area Macro Variables

Time Series Plot: DE1Y



(a) German 1-year government bond yield

Luxembourg CPI inflation



(b) Luxembourg CPI (level)

Figure 4: DE1Y and Luxembourg CPI over the sample period

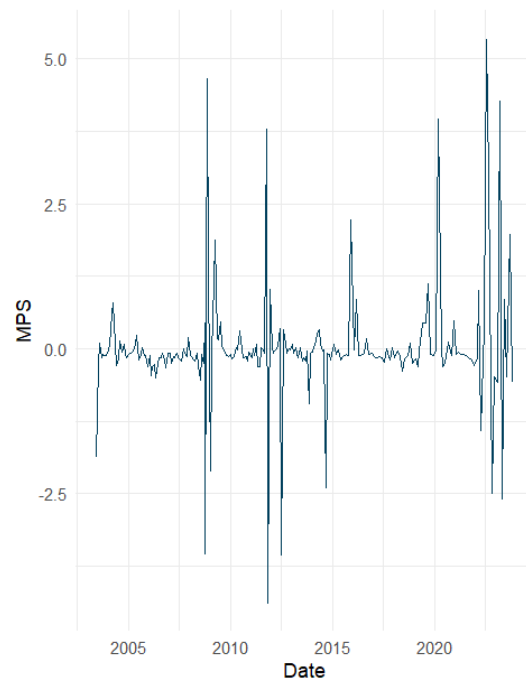


Figure 5: High-Frequency Monetary Policy Shock (Standardized PC1)

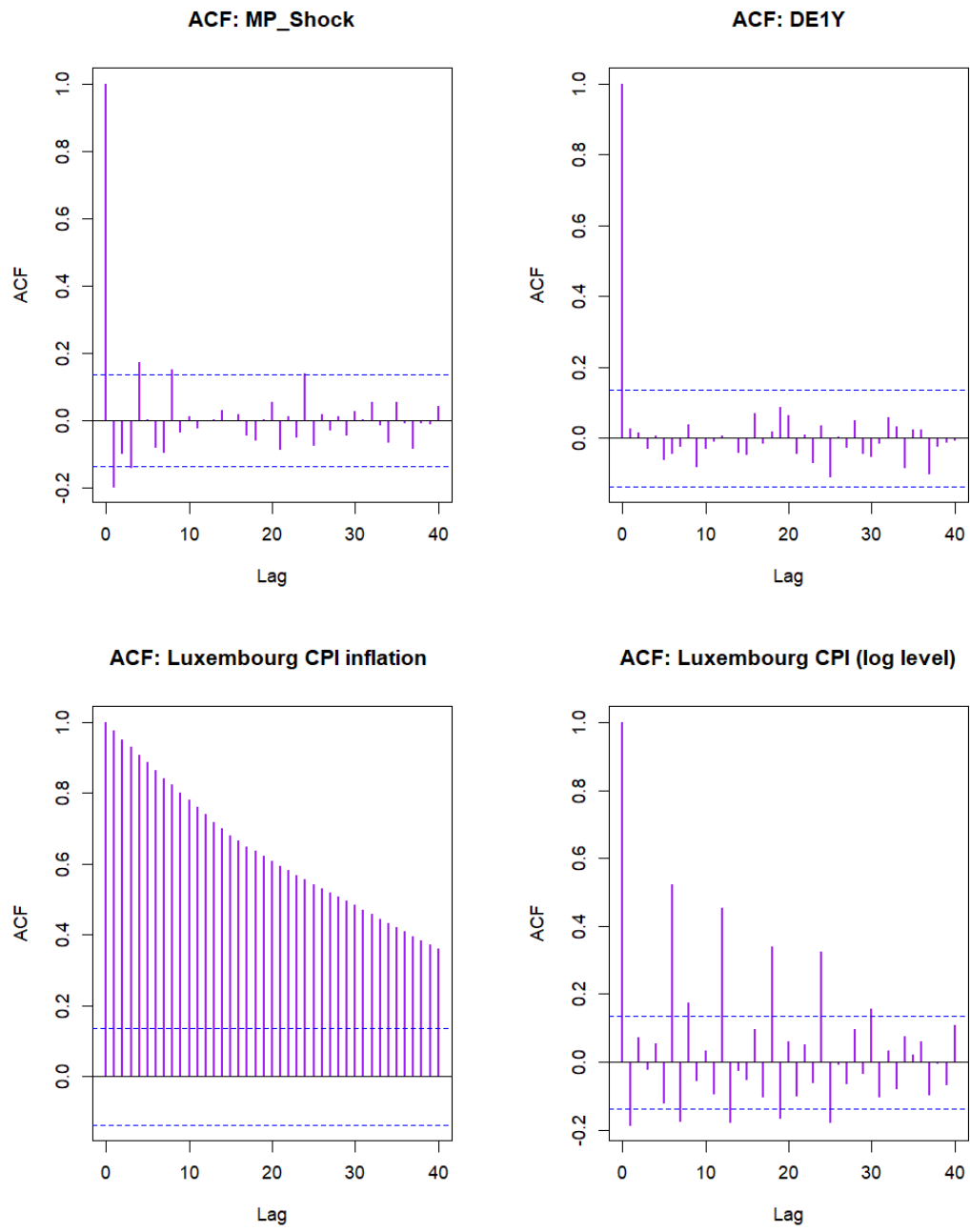


Figure 6: Autocorrelation Functions (ACF) for Key Time Series

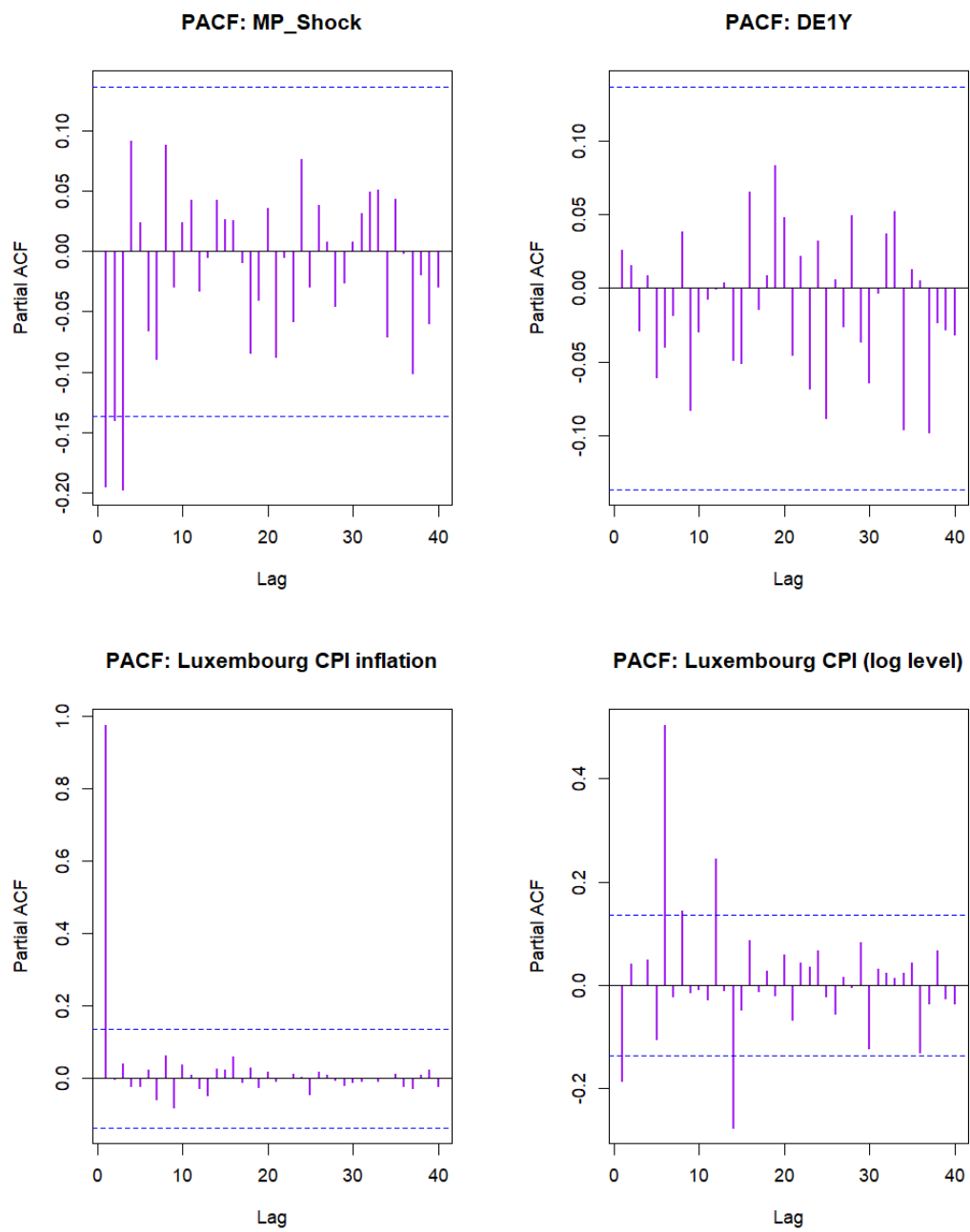


Figure 7: Partial Autocorrelation Functions (PACF) for Key Time Series

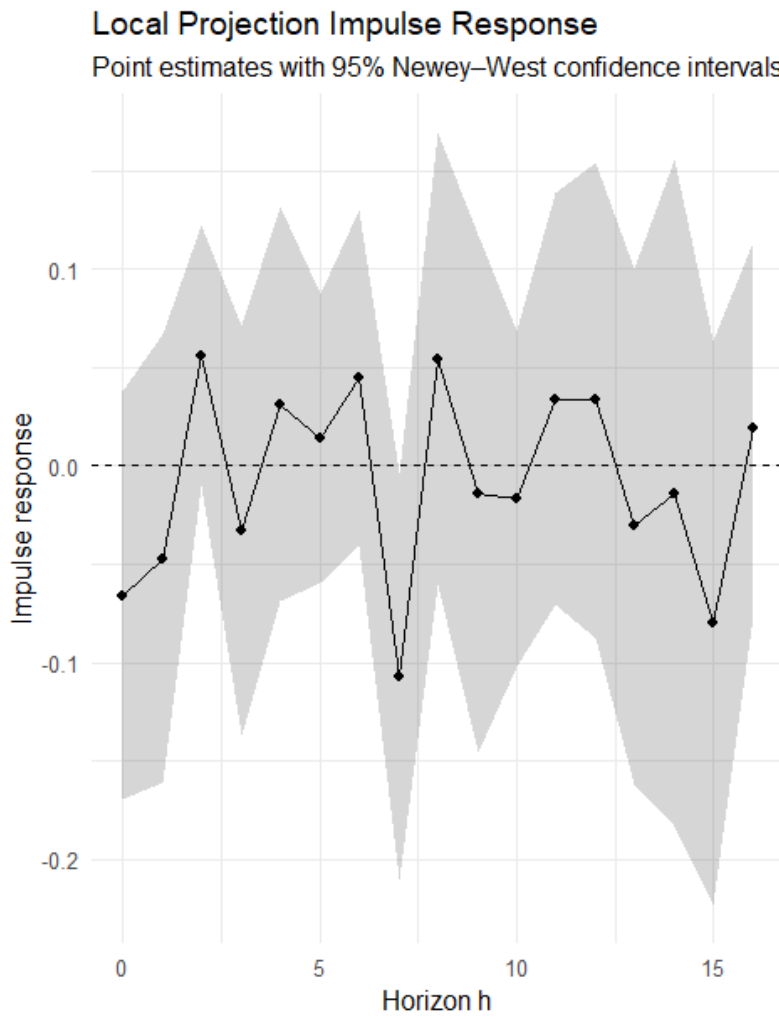


Figure 8: Local projection impulse response of Luxembourg CPI inflation to the cleaned monetary policy shock. The solid line shows the horizon-specific point estimates $\hat{\beta}_1^{(h)}$, and the shaded area the associated 95% Newey–West confidence band.

References

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¹Data sources: EA-MPD (Altavilla et al., 2019) for OIS rates and DE1Y, ECB Data Portal for STOXX, VSTOXX and industrial production, Eurostat for EA HICP, STATEC Luxembourg for Luxembourg CPI, and Yahoo Finance for the S&P 500 index.